

2. A Company has a sales of ₹ 2,00,000. the variable cost are 40% of sales. & Fixed expenses are ₹ 60,000. the interest on borrowed Capital ₹ 20,000. Compute the leverages and show the impact on taxable income when sales ↑ by 10%.

Particulars.	Amount ₹	Sales ↑ by 10%
Sales	200000	220000
less: Variable Cost (200000 × 40%)	80000	88000
Contribution	120000	132000
less: Fixed Cost	60000	60000
EBIT	60000	72000
less: Interest	20000	20000
EBT	40000	52000

$$1) \text{ Financial leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{60000}{40000} = 1.5$$

$$= \frac{72000}{52000} = 1.38$$

$$2) \text{ Operating leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{120000}{60000} = 2$$

$$= \frac{132000}{72000} = 1.83$$

$$3) \text{ Combine leverage} = \text{OL} \times \text{FL} = 1.5 \times 2 = 3$$

$$= 1.83 \times 1.38 = 2.52$$

3. From the following calculate financial, operating & combine leverage.

Sales 10000 units at ₹ 25 per unit. has selling price, variable cost ₹ 5 per unit, fixed cost ₹ 30000 and fixed interest cost ₹ 15000.

Particulars.	Amount ₹ =
Sales (10000 x 25)	250000
less: variable cost (10000 x 5)	50000
Contribution	200000
less: fixed cost	30000
EBIT	170000
less: interest	15000
EBT	155000

$$1) \text{ Financial leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{170000}{155000} = 1.09$$

$$2) \text{ Operating leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{200000}{170000} = 1.17$$

$$3) \text{ Combine leverage} = \text{OL} \times \text{FL} = 1.17 \times 1.09 = 1.2753$$

4. Evaluate 2 Companies in terms of its financial & operating lever

	A	B
Sales	2000000	3000000
Variable Cost	40% on Sale	30% on Sale
Fixed Cost	500000	700000
Interest	100000	125000

=>	Particulars	A	B
	Sales	2000000	3000000
	less: Variable Cost $\left(\begin{matrix} 40\% \times 2000000 \\ 30\% \times 3000000 \end{matrix} \right)$	800000	900000
	Contribution $\rightarrow$	1200000	2100000
	less: Fixed Cost	500000	700000
	EBIT	700000	1400000
	less: Interest	100000	125000
	EBT	600000	1275000
	Financial leverage = $\frac{EBIT}{EBT}$	$\frac{700000}{600000} = 1.166$	$\frac{1400000}{1275000} = 1.098$
	Operating leverage = $\frac{Contribution}{EBIT}$	$\frac{1200000}{700000} = 1.714$	$\frac{2100000}{1400000} = 1.50$

Firm A has greater risk in financial and business than firm B.

The following information is available in respect of a Product.

Unit sold 180000

units sales price ₹ 5

fixed cost 240000

variable cost per unit ₹ 1

Tax rate 50%

10% Debt Capital 600000

Calculate operating leverage, Financial leverage & combine leverage.

Particulars

Amount
₹

Sales (180000 × 5)

900000

less : variable cost (180000 × 1)

180000

Contribution

720000

less : fixed cost

240000

EBIT

480000

less : Interest (600000 × 10%)

60000

EBT

420000

less : Tax 50% (420000 × 50%)

210000

EAT

210000

$$\text{Financial leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{480000}{420000} = 1.14$$

$$\text{Operating leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{720000}{480000} = 1.5$$

$$\text{Combine leverage} = \text{OL} \times \text{FL} = 1.5 \times 1.14 = 1.71$$

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6. Balance sheet of Xc limited on 31/3/2020 has follows

Liabilities	Amount	Assets	Amount
Eq. Capital (10 per share)	60000	Fixed assets	150000
10% Debentures	80000	Current assets	50000
Retain earnings	20000		
Current liabilities	40000		
	<u>200000</u>		<u>200000</u>

The Companies turnover share is 3. Its fixed cost is ₹ 100000 & variable cost is 40%. The tax rate is 50%.

A. Calculate all the 3 types of leverages.

B. Determine the likely level of EBIT if EPS is ₹ 5.

⇒ Particulars Amount

Total assets turnover ratio = $\frac{\text{Sales}}{\text{Total assets}}$

$$3 = \frac{\text{Sales}}{200000}$$

$$\text{Sales} = 200000 \times 3$$

$$= 600000$$

Particulars	Amount
Sales	600000
less: variable Cost $(600000 \times 40\%)$	240000
Contribution	360000
less: Fixed Cost	100000
EBIT	260000
less: Interest $(80000 \times 10\%)$	8000
EBT	252000
less: Tax $(50\% \times 252000)$	126000
EAT	126000

Combine leverage = $DL \times FL = 1.38 \times 1.03 = 1.426$

7. Determine the earning per share of a Co. which has Operating Profit (EBIT) of ₹ 160000. its Capital structure consist of following securities.
10% debentures 500000, 12% Preference shares 100000.
Eq. sh of 100 each. 400000.

1) The Co. is in the 55% tax. determine the Co EPS.

2) Determine the percentage change in EPS. associated with 30% ↑ & 30% ↓

		30% ↑ in EBIT	30% ↓ in EBIT
EBIT	160000	208000	112000
less: Interest (10% P500000)	50000	50000	50000
EBT	110000	158000	62000
less: Tax (55% × 110000)	60500	86900	34100
EAT	49500	71100	27900
less: Preference dividend (100000 × 12%)	12000	12000	12000
Earning Available ESH	37500	59100	15900

$$\text{EPS} = \frac{37500}{4000} = 9.375$$

$$\text{EPS} = \frac{59100}{4000} = 14.8$$

$$\text{EPS} = \frac{15900}{4000} = 3.97$$

2) when the EPS, EBIT increasing by 30%.

$$= \frac{\text{EPS @ 30\% increase} - \text{EPS @ 160000}}{\text{EPS @ 160000}} \times 100$$

$$= \frac{14.8 - 9.4}{9.4} \times 100$$

$$= 57.44$$

3) When the EPS, EBIT Decreased by 30%

$$= \frac{\text{EPS @ 160000} - \text{EPS @ 30\% ↓}}{\text{EPS @ 160000}} \times 100 = \frac{9.4 - 3.9}{9.4} \times 100 = 58.51$$

$$\text{Financial leverage} = \frac{\text{EBIT}}{\text{EBT}}$$

$$= \frac{160000}{110000} = \frac{208000}{158000} = \frac{118000}{68000}$$

$$= 1.45 = 1.31 = 1.80$$

A Private Co has EBIT of ₹ 480000 and its Capital Structure Consists of following securities.

Eq. sh Capital ₹ 100 each 400000.

12% Pref. sh 600000.

14.5% Debentures 1000000.

The Co is facing fluctuation in its sales wt would be the changing EPS.

1) If EBIT ↑ by 25% and ↓ by 25%. Determine the Percentage changes in EPS if ↑ @ ↓ by 25%.

2) Corporate tax 35%.

	EBIT @ 480000	↑ by 25%	↓ by 25%
EBIT	480000	600000	360000
less: Interest (1000000 × 5%)	145000	145000	145000
EBT	335000	455000	215000
less: Tax 35%	150500	192500	108500
EAT	279500	357500	201500
less: Preference dividends (600000 × 12%)	87000	87000	87000
Earnings Available to Eq. sh	192500	270500	114500
EPS = $\frac{\text{EA to Eq. sh holder}}{\text{no of shares}}$	$\frac{192500}{4000}$	$\frac{270500}{4000}$	$\frac{114500}{4000}$
no of shares = $\frac{400000}{100}$	48.125	67.625	28.625

= 4000

when EPS, EBIT ↑ by 25%

$$= \frac{\text{EPS @ } \uparrow 25\% - \text{EPS @ } 480000}{\text{EPS @ } 480000} \times 100$$

$$= \frac{67.625 - 48.125}{48.125} \times 100$$

$$= 40.5194$$

when EPS, EBIT ↓ by 25%

$$= \frac{\text{EPS @ } 480000 - \text{EPS @ } \downarrow 25\%}{\text{EPS @ } 480000} \times 100$$

$$= \frac{48.125 - 28.625}{48.125} \times 100$$

$$= 40.5194$$

17/7/23

9. Find the Financial leverage & operating leverage under situation A & B. from the following financial Plan 1 & 2 respectively, from the following related to operations and Capital structure of ABC Ltd.
Installed Capacity 1000 units.

Actual Production & sales 800 units.

Selling price per unit ₹ 20.

Variable Cost per unit ₹ 15.

Fixed Cost

Situation A ₹ 200, Situation B ₹ 1500.

Capital structured :-

Particulars

Eq. Capital

Debt

Cost of Debt 10%

Plan 1 / 200A Plan 2

5000

7000

5000

2000

* when situation A Fixed cost is 800

	Plan 1	Plan 2
Sales (800×20)	16000	16000
less: variable cost (800×15)	12000	12000
Contribution	4000	4000
less: Fixed cost	800	800
EBIT	3200	3200
less: Interest $(5000 \times 10\%)$ $(2000 \times 10\%)$	500	200
EBT	2700	3000
Financial leverage = $\frac{EBIT}{EBT}$	$\frac{3200}{2700}$	$\frac{3200}{3000}$
	$= 1.185$	$= 1.066$
Operating leverage = $\frac{Contribution}{EBIT}$	$\frac{4000}{3200}$	$\frac{4000}{3200}$
	$= 1.25$	$= 1.25$

* when situation B Fixed cost is 1500

Sales (800×20)	16000	16000
less: variable cost (800×15)	12000	12000
Contribution	4000	4000
less: Fixed cost	1500	1500
EBIT	2500	2500
less: Interest	500	200
EBT	2000	2300
Financial leverage = $\frac{EBIT}{EBT}$	$\frac{2500}{2000}$	$\frac{2500}{2300}$
	$= 1.25$	$= 1.086$
Operating leverage = $\frac{Contribution}{EBIT}$	$\frac{4000}{2500}$	$\frac{4000}{2500}$
	$= 1.6$	$= 1.6$

10 A Co. Ltd has a Capital of ₹ 100000 divided into sh of ₹ 10 each. It has major expansion program requiring an investment of another ₹ 50000. The management is considering the following alternatives for raising this amount.

Issue of 5000 shares ₹ 10 each.

Issue of 5000 12% Preference shares ₹ 10 each.

Issue of 10% Debentures ₹ 50000.

The Company's (Present) earnings before interest & tax (EBIT) ₹ 30000 per annum. You are required to calculate the effect of each of the above modes of financial on earning per share.

Presuming 1) EBIT continued to be same after expansion

2) EBIT ↑ by ₹ 10000.

3) Tax liability 50%.

⇒	Eq. sh of 5000	12% Pref. sh of 5000	10% Deb of 50000
EBIT	30000	30000	30000
less: Interest	—	—	5000
EBT	30000	30000	25000
less: Tax 50%	15000	15000	12500
EAT	15000	15000	12500
less: Dividend	—	6000	—
Earning Available ESH	15000	9000	12500
$EPS = \frac{EA \text{ Eq. sh. hol}}{\text{no. of Eq. sh}}$	$= \frac{15000}{5000}$	$= \frac{9000}{10000}$	$= \frac{12500}{10000}$
$Eq. sh = \frac{50000}{10} = 5000$	$= 3$	$= 0.9$	$= 1.25$
Existing sh = $\frac{100000}{10}$			

	Eq sh of 50000 @ 10%	12% Ppf sh of 5000 @ 10%	10% Debentures of 50000.
EBIT	40000	40000	40000
less: Interest	—	—	5000
EBT	40000	40000	35000
less: Tax 50%	20000	20000	17500
EAT	20000	20000	17500
less: Ppf dividend	—	6000	—
E A Eq. sh. hol	20000	14000	17500
EPS = $\frac{E A \text{ Eq. sh. ho}}{\text{no of Eq. sh}}$	$\frac{20000}{5000}$	$\frac{14000}{10000}$	$\frac{17500}{10000}$
	4	1.4	1.75

3. X Ltd is capitalise with ₹ 100000 divided into 100000 Eq sh of ₹ 10 each. The management decided to raise another 1000000 to finance the major expansion programme. There are 4 possible financial Plan

- 1) All Eq. sh.
- 2) ₹ 500000 in Eq and balanced in debentures carrying 8% interest
- 3) All debentures carry 8% interest.
- 4) ₹ 500000 in Eq & ₹ 500000 in Ppf sh carrying 10% dividend.

The existing EBIT amount ₹ 120000 per annum.

- 1) Calculate EPS in all the above 4 plans.
- 2) Calculate EPS if EBIT is double due to expansion program.

If the tax rate has not given we should assume that 50%.

	Eq. sh 1000000	5L Pn Eq bal Deb @ 10%	8% Deb bank	50% Eq 5L P-sh 10%
EBIT	120000	120000	120000	120000
less: Interest	—	50000	80000	—
EBT	120000	70000	40000	120000
less: Tax	60000	35000	20000	60000
EAT	60000	35000	20000	60000
less: Pay dividend	—	—	—	50000
EA Eq. sh. holders	60000	35000	20000	10000
EPS = $\frac{EA}{\text{Eq. sh}}$	$\frac{60000}{100000}$	$\frac{35000}{50000}$	$\frac{20000}{100000}$	$\frac{10000}{50000}$
no of Eq. sh = $\frac{1000000}{10}$	= 0.6	= 0.7	= 0.2	= 0.1
	= 100000			
$\frac{500000}{10} = 50000$				

when EBIT double due to expansion: 240000

	Eq. sh. 1000000	5L Pn Eq & 10% Deb bal	8% Deb	5L Eq. sh 5L P-sh 10%
EBIT	240000	240000	240000	240000
less: Interest	—	50000	80000	—
EBT	240000	190000	160000	240000
less: Tax 50%	120000	95000	80000	120000
EAT	120000	95000	80000	120000
less: Pay dividend	—	—	—	50000
EA. Eq. sh holders	120000	95000	80000	70000
EPS = $\frac{EA}{\text{Eq. sh}}$	$\frac{120000}{100000}$	$\frac{95000}{50000}$	$\frac{80000}{100000}$	$\frac{70000}{50000}$
	1.2	1.9	0.8	1.4

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Anand Hld has currently on all Eq. structure consisting of 10000 Eq.sh of ₹ 100 each.

The management is planning to raise another ₹ 2000000 to finance major expansion programme and he considering 3 alternative method of financing.

1) To issue 20000 Eq.sh of 100 each

2) To issue 20000 7% Debenture ₹ 100 each.

3) To issue 20000 8% Pfd.sh 100 each.

Anand Hld is expecting operating Profit 1200000. The tax Rate is 42% Determine EPS. In each alternatives and Comment which alternative is suitable.

Particulars	Eq.sh 20000 @ 100	20000 7% Deb @ 100	20000 8% Pfd.sh 100.
EBIT	1200000	1200000	1200000
less: Interest	—	140000	—
EBT	1200000	1060000	1200000
less: Tax 42%	504000	445200	504000
EAT	696000	614800	696000
less: Pfd dividend	—	—	160000
EAT ESH	696000	614800	536000
EPS = $\frac{EAT ESH}{\text{no of sh}}$	$\frac{696000}{20000}$	$\frac{614800}{20000}$	$\frac{536000}{20000}$
	34.8	30.74 61.48	26.8 53.6

We recommend situation 2 due to highest dividend & financially consistent.

14 Following are financial data for Companies A, B & C.
Prepare Income Stmt.

Particulars	A	B	C
Variable cost has % of sales	66% (2/3)	75%	50%
Interest expenses	200	300	1000
Degree of OL	5-1	6-1	2-1
FL	3-1	4-1	2-1
Income tax rate	50%	50%	50%

$$\Rightarrow \text{Degree of FL} = \frac{\text{EBIT}}{\text{EBIT}}$$

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBIT}}$$

$$3 = \frac{\text{EBIT}}{\text{EBIT} - 200}$$

$$4 = \frac{\text{EBIT}}{\text{EBIT} - 300}$$

$$3\text{EBIT} - 600 = \text{EBIT}$$

$$4\text{EBIT} - 1200 = \text{EBIT}$$

$$2\text{EBIT} = 600$$

$$3\text{EBIT} = 1200$$

$$\text{EBIT} = \frac{600}{2}$$

$$\text{EBIT} = \frac{1200}{3}$$

$$\text{EBIT} = 300$$

$$\text{EBIT} = 400$$

$$\text{DOL} = \frac{\text{EBIT}}{\text{EBIT}}$$

$$\text{DOL} = \frac{C}{\text{EBIT}}$$

$$2 = \frac{\text{EBIT}}{\text{EBIT} - 1000}$$

$$5 = \frac{C}{300}$$

$$2\text{EBIT} - 2000 = \text{EBIT}$$

$$5 \times 300 = C$$

$$\text{EBIT} = 2000$$

$$1500 = C$$

$$\text{DOL} = \frac{C}{\text{EBIT}}$$

$$\text{DOL} = \frac{C}{\text{EBIT}}$$

$$6 = \frac{C}{400}$$

$$2 = \frac{C}{2000}$$

$$C = 6 \times 400$$

$$2 \times 2000 = C$$

$$C = 2400$$

$$4000 = C$$

If no information about sales. Consider that has 100

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$$\text{Contribution} = \text{Sales} - \text{Variable Cost}$$

$$A = 100 - 66.67$$

$$= 33.33\%$$

$$B = 100 - 75$$

$$= 25\%$$

$$C = 100 - 50$$

$$= 50\%$$

Calculation of Sales

If Contribution is (£1500) 33.33%, (£2400) 25%, (£4000) 50%
what is the sales.

$$A = \frac{1500}{33.33} \times 100 = 4500$$

$$B = \frac{2400}{25} \times 100 = 9600$$

$$C = \frac{4000}{50} \times 100 = 8000$$

Income stmt

Particulars	A	B	C
Sales	4500	9600	8000
less: variable	7500	7200	4000
Contribution	1500	2400	4000
less: Fixed Cost	1200	2000	2000
EBIT	300	400	2000
less: Interest	200	300	1000
EBT	100	100	1000
less: Tax 50%	50	50	500
	50	50	500